

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	Y. Varun Kumar Reddy	Reg. No.:	192424230
Section / Batch:	2024	Date:	13/07/2026

Code of Conduct

I_Y. Varun Kumar Reddy_(192424230) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Develop a modular and efficient Python program that satisfies all the functional requirements specified in the problem statement.
- Demonstrate the correctness of the implementation using suitable test cases and justify the output obtained.
- Follow good programming practices, including meaningful variable names, proper code indentation, comments, and error handling wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Information Extraction	Regex-Based Information Extraction	1
Pattern Matching	Regex Pattern Matching	2
Regular Expression Patterns	Regex-Based Input Validation	3

Section 1: Regular Expressions – Resume Information Extraction

1. A multinational recruitment company receives thousands of resumes every day in text format. The HR department requires an automated system to extract candidate information for shortlisting.

Design and implement a Python-based resume information extraction system using Regular Expressions that performs the following tasks:

Extract the candidate's name.

Identify email addresses.

Extract mobile numbers.

Detect technical skills (Python, Java, SQL, Machine Learning, NLP).

Extract years of experience.

Generate a structured summary of the candidate's profile.

Display candidates who satisfy the minimum eligibility criteria (minimum 2 years of experience and Python skill).

Aim

Develop a Python program using Regular Expressions (Regex) to extract important details from a candidate's resume and identify eligible candidates.

Description

This program automates the resume screening process. It reads resume text and uses Regular Expressions to extract the candidate's name, email address, mobile number, technical skills, and years of experience. After extracting the information, it generates a structured summary and checks whether the candidate satisfies the minimum eligibility criteria of having at least 2 years of experience and Python as one of the technical skills.

Modules Used

- re (Regular Expression module)

Functions Used

- re.search() – Finds the first matching pattern.
- re.findall() – Finds all matching patterns.
- re.IGNORECASE – Performs case-insensitive matching.

Expected Output

- Candidate Name
- Email Address
- Mobile Number
- Technical Skills
- Years of Experience
- Eligibility Status (Eligible / Not Eligible)

CODE :

```
import re

resume = """
Name: Varun Reddy
Email: varunreddy@gmail.com
Mobile: 9876543210
Skills: Python, Java, SQL, Machine Learning, NLP
Experience: 3 years
"""

name = re.search(r"Name:\s*(.*)", resume)
```

```

if name:
    name = name.group(1)

emails = re.findall(r"[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}", resume)

mobile = re.findall(r"\b[6-9]\d{9}\b", resume)

skills = ["Python", "Java", "SQL", "Machine Learning", "NLP"]
found_skills = []

for skill in skills:
    if re.search(skill, resume, re.IGNORECASE):
        found_skills.append(skill)

exp = re.search(r"(\d+)\s+years", resume)
experience = int(exp.group(1)) if exp else 0

print("Candidate Summary")
print("-----")
print("Name:", name)
print("Email:", emails)
print("Mobile:", mobile)
print("Skills:", found_skills)
print("Experience:", experience, "years")

if experience >= 2 and "Python" in found_skills:
    print("Eligible for Shortlisting")
else:
    print("Not Eligible")

```

OUTPUT :

```

PS C:\Users\reddy\nlp> & 'C:\Users\reddy\AppData\Local\Python\pythoncore-3.14-64\python\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '51844' '--' 'C:\Users\reddy\nlp\resume.txt'
Candidate Summary
-----
Name: Varun Reddy
Email: ['varunreddy@gmail.com']
Mobile: ['9876543210']
Skills: ['Python', 'Java', 'SQL', 'Machine Learning', 'NLP']
Experience: 3 years

```

Section 2: Regular Expressions – Product Search System

2. An e-commerce company plans to enhance its product search engine to improve customer experience through flexible keyword matching.

Design and implement a Python-based product search system using Regular Expressions that performs the following tasks:

Search products using an exact keyword.

Perform prefix-based searches.

Perform suffix-based searches.

Search using partial keywords.

Perform case-insensitive searches.

Display all matching product names.

Generate a report showing the total number of matching products for each search.

Aim

Develop a Python-based product search system using Regular Expressions to perform flexible product searches.

Description

This program simulates an e-commerce product search engine. It stores a list of product names and allows users to search using different techniques such as exact match, prefix search, suffix search, partial keyword search, and case-insensitive search. It then displays all matching products and generates a report showing the total number of matching products.

Modules Used

- re (Regular Expression module)

Functions Used

- re.fullmatch() – Performs exact matching.
- re.match() – Performs prefix matching.
- re.search() – Performs suffix and partial matching.
- re.IGNORECASE – Makes the search case-insensitive.

Expected Output

- Exact Matching Products
- Prefix Matching Products
- Suffix Matching Products
- Partial Matching Products
- Case-Insensitive Matching Products
- Total Number of Matching Products

CODE :

```
import re
```

```
products = [  
    "Laptop",  
    "Laptop Bag",  
    "Gaming Mouse",  
    "Bluetooth Speaker",  
    "Smart Phone",  
    "Phone Charger",  
    "Wireless Keyboard",  
    "USB Cable"  
]
```

```
keyword = input("Enter search keyword: ")

print("\nExact Match")
for p in products:
    if re.fullmatch(keyword, p, re.IGNORECASE):
        print(p)

print("\nPrefix Match")
for p in products:
    if re.match(keyword, p, re.IGNORECASE):
        print(p)

print("\nSuffix Match")
for p in products:
    if re.search(keyword + r"$", p, re.IGNORECASE):
        print(p)

print("\nPartial Match")
count = 0
for p in products:
    if re.search(keyword, p, re.IGNORECASE):
        print(p)
        count += 1

print("\nCase-Insensitive Match")
for p in products:
    if re.search(keyword, p, re.IGNORECASE):
        print(p)

print("\nReport")
print("Total Matching Products =", count)
```

OUTPUT :

```

PS C:\Users\reddy\nlp> c:; cd 'c:\Users\reddy\nlp'; & 'C:\Users\reddy\AppData\Local\Python\python\Users\reddy\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcherdy\nlp\Q2.py'
Enter search keyword: phone

Exact Match

Prefix Match
Phone Charger

Suffix Match
Smart Phone

Partial Match
Smart Phone
Phone Charger

Case-Insensitive Match
Smart Phone
Phone Charger

Report
Total Matching Products = 2

```

Section 3: Regular Expressions – University Registration System

A university is developing an automated online registration portal to verify student information before course enrollment.

Design and implement a Python-based validation system using Regular Expressions that performs the following tasks:

- Validate the student register number.
- Validate the institutional email address.
- Validate the course code.
- Validate the semester information.
- Validate the student's mobile number.
- Display appropriate validation messages for each field.
- Generate a final registration status report indicating whether the registration is successful.

Aim

Develop a Python program using Regular Expressions to validate student registration details.

Description

This program validates the information entered by a student during university registration. It checks the register number, institutional email address, course code, semester, and mobile number against predefined Regular Expression patterns. If all fields are valid, the program displays "Registration Successful"; otherwise, it displays "Registration Failed" along with appropriate validation messages.

Modules Used

- re (Regular Expression module)

Functions Used

- re.fullmatch() – Validates the complete input against the required pattern.

Expected Output

- Register Number Validation
- Email Validation
- Course Code Validation
- Semester Validation
- Mobile Number Validation

- Final Registration Status

CODE :

```
import re

reg = input("Enter Register Number: ")
email = input("Enter College Email: ")
course = input("Enter Course Code: ")
semester = input("Enter Semester: ")
mobile = input("Enter Mobile Number: ")

status = True

# Register Number
if re.fullmatch(r"\d{2}[A-Z]{3}\d{4}", reg):
    print("Register Number Valid")
else:
    print("Invalid Register Number")
    status = False

# Institutional Email
if re.fullmatch(r"[a-zA-Z0-9._%+-]+@saveetha\.com", email):
    print("Email Valid")
else:
    print("Invalid Email")
    status = False

# Course Code
if re.fullmatch(r"[A-Z]{3}\d{2}", course):
    print("Course Code Valid")
else:
    print("Invalid Course Code")
    status = False

# Semester
if re.fullmatch(r"[1-8]", semester):
    print("Semester Valid")
else:
    print("Invalid Semester")
    status = False

# Mobile Number
if re.fullmatch(r"[6-9]\d{9}", mobile):
    print("Mobile Number Valid")
else:
    print("Invalid Mobile Number")
    status = False

# Final Report
print("\nRegistration Status")
if status:
    print("Registration Successful")
else:
```

```
print("Registration Failed")
```

OUTPUT :

```
PS C:\Users\reddy\nlp> c::; cd 'c:\Users\reddy\nlp'; & 'C:\Users\reddy\A
\Users\reddy\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bun
dy\nlp\Q3.PY'
Enter Register Number: 23ADR1234
Enter College Email: varun@saveetha.com
Enter Course Code: DSA03
Enter Semester: 5
Enter Mobile Number: 9876543210
Register Number Valid
Email Valid
Course Code Valid
Semester Valid
Mobile Number Valid

Registration Status
Registration Successful
PS C:\Users\reddy\nlp>
```


Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q. No. / Criteria	Understanding of Problem Context	Application of Concepts	Problem-Solving Approach	Accuracy of Solution	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____